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Review

The Diseases and Pests of Rubber Tree and Their Natural Control Potential: A Bibliometric Analysis

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Abstract: In order to trace the research history of diseases and pests in rubber tree and explore the potential for their natural control, a bibliometric analysis was conducted based on relevant documents retrieved from the Clarivate Analytics Web of Science (WoS) core collection SCI-E database. VOSviewer software was utilized to analyze the research distribution, scientific collaboration, knowledge structure, and research frontiers. The results show that annual publications on the diseases and pests of rubber tree have increased rapidly after 2005 after a long period of emergence and fluctuation. A total of 624 relevant publications from 51 countries/regions were identified. China was the most productive country with 152 documents, most of which were related to Colletotrichum leaf disease, powdery mildew, and other emerging diseases of rubber tree. France and Brazil produced rich research to tackle South American leaf blight, and have established a close collaborative relationship. Based on the analysis of themes and trend topics, pathogenicity mechanisms of fungal pathogens and plant defense mechanisms are currently hot topics. By further looking into the research, the defense-related genes of rubber tree and antagonistic mechanisms behind candidate biocontrol agents reveal great potential in developing natural control strategies. This study provides a useful reference about the progress and evolution of research into diseases and pests in rubber tree.

Keywords: diseases and pests; rubber tree; natural control potential; bibliometric analysis

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Citation: Chen, L.; Xu, L.; Li, X.; Wang, Y.; Feng, Y.; Huang, G. The Diseases and Pests of Rubber Tree and Their Natural Control Potential: A Bibliometric Analysis. *Agronomy* **2023**, *13*, 1965. https://doi.org/10.3390/agronomy13081965

Academic Editor: Luca Ruiu

Received: 31 May 2023
Revised: 18 July 2023
Accepted: 21 July 2023
Published: 25 July 2023

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1. Introduction

Rubber tree (*Hevea brasiliensis*), a perennial crop originating from the Amazon, has been widely cultivated across the tropical belt in order to supply the global demand for rubber [1]. Although rubber tree contributes to the welfare of farmers and has found its place in national economies of many rubber-producing countries, its cultivation is impeded by severe diseases and harmful pests [2]. Particular fungal diseases inhibit the production of natural rubber on a commercial scale. One of the most severe diseases in the rubber plantations of South and Central America is South American leaf blight, which is caused by the ascomycete *Microcyclus ulei* (later renamed *Pseudocercospora ulei*). Although the disease distribution is continentally restricted, its potential to spread worldwide is rising as more transcontinental airlines directly connect tropical countries [3]. Other leaf fungal diseases that seriously affect rubber growing regions of tropical countries are caused by pathogens such as *Phytophthora* species, *Colletotrichum* species, and *Corynespora cassiicola*. For example, *Colletotrichum gloeosporioides* and *C. acutatum* are reported as major causal agents of rubber leaf anthracnose in several countries, including Sri Lanka and China [4,5], and *C. cassiicola* has been proven responsible for leaf fall disease of rubber tree in all rubber producing countries in Africa and Asia [6]. In addition to leaf diseases, root diseases of rubber tree caused by certain soilborne pathogens such as *Rigidoporus microporus* pose great threat to the rubber industry [7]. In addition to the above diseases, pests inhabiting rubber

plantation, for example, phytophagous mites, hornworm and lace bug have been reported to cause great threat [8,9].

The need to explore natural control strategies against these rubber diseases is urgent, and applications of comprehensive control measures have proceeded accordingly [10]. Although efforts to control rubber diseases have been taken since 1910, rubber farmers continue to suffer a great deal of crop failure [1]. During this long history of investigation, researchers have proposed various disease control strategies, for example, the use of modern chemical fungicides, breeding of rubber resistant clones, application of biocontrol agents, and searching for growing zones suitable for rubber and with adverse climatic condition for disease development [10–13]. However, how to integrate these control strategies to achieve upgraded and improved disease control effectiveness remains a challenge.

Due to the abundance of information on diseases and pests of rubber tree, a bibliometric review can be useful for following the research progress and summarizing the current research topics [14,15]. In order to analyze the scientific literature regarding diseases and pests of rubber tree, we use bibliometric methods and carry out a quantitative study based on relevant publications extracted from the Web of Science-SCIE database. We point out the most productive countries/organizations/authors, trace the overall research development over time, and identify the major research topics. We aim to provide an overview of the diseases and pests of rubber tree and present a quick reference guide for researchers. Furthermore, we highlight disease control studies and try to explore their natural control potential.

2. Materials and Methods

2.1. Data Collection

This study focuses on a bibliometric analysis of scientific publications related to the diseases and pests of rubber tree. The relevant publications were retrieved from the Clarivate Analytics Web of Science (WoS) core collection SCIE database. The SCIE database, as an international scientific literature database, allows publications to be searched using both advanced and basic methods [16]. The retrieval time spanned from 1 January 1900 to 13 July 2023. The following Boolean strings were used: ((“rubber wood*” OR “rubber seedling*” OR “rubber tree*” OR “rubber plant*” OR Hevea) AND (disease* OR spot OR rot OR fall* OR anthracnose OR powdery OR blight OR canker OR *pathogen* OR *bacteria* OR *fung* OR microb* OR pest* OR *mite* OR insect* OR worm* OR nematode* OR bug OR infect* OR infest* OR *parasiti* OR resistan* OR defen* OR *control* OR antagonist* OR agent*)) (Topic) or (rubber NEAR (blight OR spot OR disease* OR canker)) (Topic) or (Hevea NEAR (blight OR spot Or disease*)) (Topic) or ((Erysiphe OR Corynespora OR Colletotrichum OR Rigidoporus OR Oidium OR ulei OR Phytophthora OR Ascomyc* OR “novel species” OR “sp. nov.” OR “new species” OR Pythium OR calcarus OR “First report” OR Acari OR Meloidogyne OR erinyis OR tenuipalpid) AND (rubber* OR Hevea*)) (Topic). Searching within the topic field provides papers that contain the defined strings in a publication’s title, abstract, author keywords, and keywords plus; “*” indicates any groups of characters, including blank characters. The Boolean operators were used to detect document with various keyword combinations. Duplicates were removed using Excel. Irrelevant studies were manually excluded by reading the abstracts and titles of papers. After artificial screening, a total of 624 papers were retrieved for further study, as shown in Table 1; after this selection process, all the selected international and domestic literature was in the category of infectious diseases/pests of rubber trees. The flowsheet of the document collection process is illustrated in Figure 1.

Table 1. Main information about document data and the distribution of the publication type in the field of diseases and pests of rubber tree.

Description	Results
Main information about data	
Timespan	1915:2023
Sources (Journals, Books, etc.)	222
Documents	624
Annual Growth Rate %	3.06
Document Average Age	17.1
Average citations per doc	10.95
References	16,229
Document Types	
Article	534
News Item	17
Note	17
Review	16
Editorial Material	13
Meeting Abstract	13
Article; Early Access	7
Article; Proceedings Paper	4
Book Review	1
Letter	1
Review; Book Chapter	1

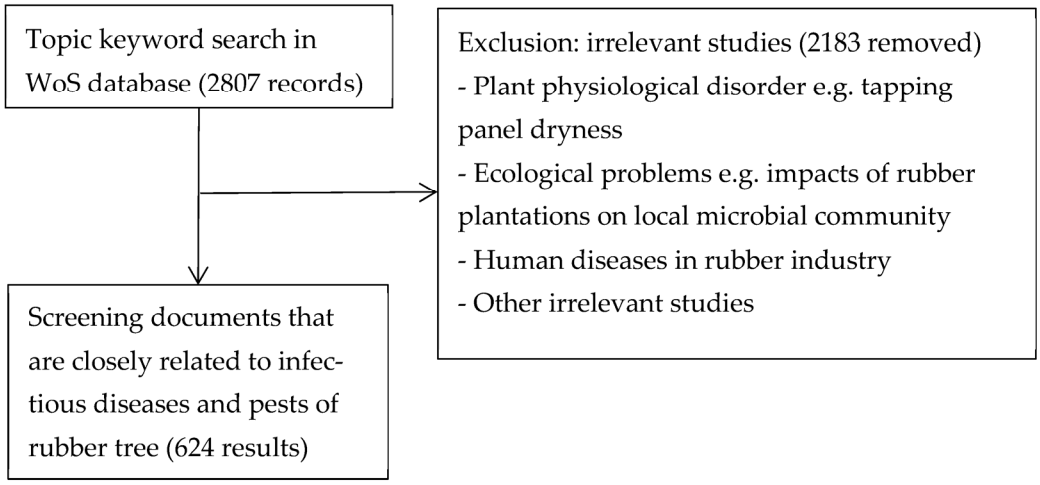


Figure 1. The flowsheet of the document collection.

2.2. Data Analysis

The information for the documents in CSV or TXT format that met the requirements for bibliometric analysis contained the document type, author, title, abstract, year of publication, journal, affiliation, country/region, keywords, and citation count. After extracting this information from the documents, the year of publication, journal, and citation count were analyzed using the Bibliometrix R package on R version 4.2.1 [17]. The H index is an important evaluation parameter that reflects the academic influence of journals. Source analysis was conducted by applying Bradford's law using Bibliometrix R package on R version 4.2.1, as in other recent works [18,19]. Scimago Graphica_1.0.23 incorporating VOSviewer software version 1.6.18 was used to map the co-authorship of countries/regions [20]. We used VOSviewer_1.6.18 software to conduct bibliometric analyses and visualize author networks, organizational networks, country networks, and the occurrence of terms that were extracted from titles and abstracts. According to Van Eck and Waltman [21–23], VOSviewer is a powerful tool for visualizing collaborative networks of authors, organizations, and countries

by using a clustering algorithm based on co-authorship. Cluster analysis of main research terms creates a term network based on their co-occurrence and allows for exploration of the most important terms for each cluster and their relations [21]. Through co-occurrence and frequency analysis of terms, the trends and emphases of research can be analyzed.

3. Results

This section aims to provide a comprehensive analysis of the scientific publications related to diseases and pests of rubber tree and examine their natural control strategies. It is made up of the following five parts: Section 3.1 provides an overview of the selected publications and the distribution of their sources; Section 3.2 presents the publication distribution and networks of authors, organizations, and countries/regions; Section 3.3 presents a bibliometric analysis of research themes and trend topics; and Sections 3.4 and 3.5 look into the publications related to resistance related-studies and screening of biocontrol agents, respectively.

3.1. Overview of the Selected Publications

As Figure 2 shows, publication volume per year dating back to the year 1915 shows the academic development in the field of diseases and pests of rubber tree. The literature retrieval time was to 13 July 2023. Generally, until the 1966 only one or two papers per year were recorded. From 1967 to 2004 there were no more than eleven publications per year, with considerable annual fluctuation. However, after 2005 significantly higher interest in these research topics is revealed by an accelerating upward growth curve; in the year 2022, the amount of documents reached a peak of 56.

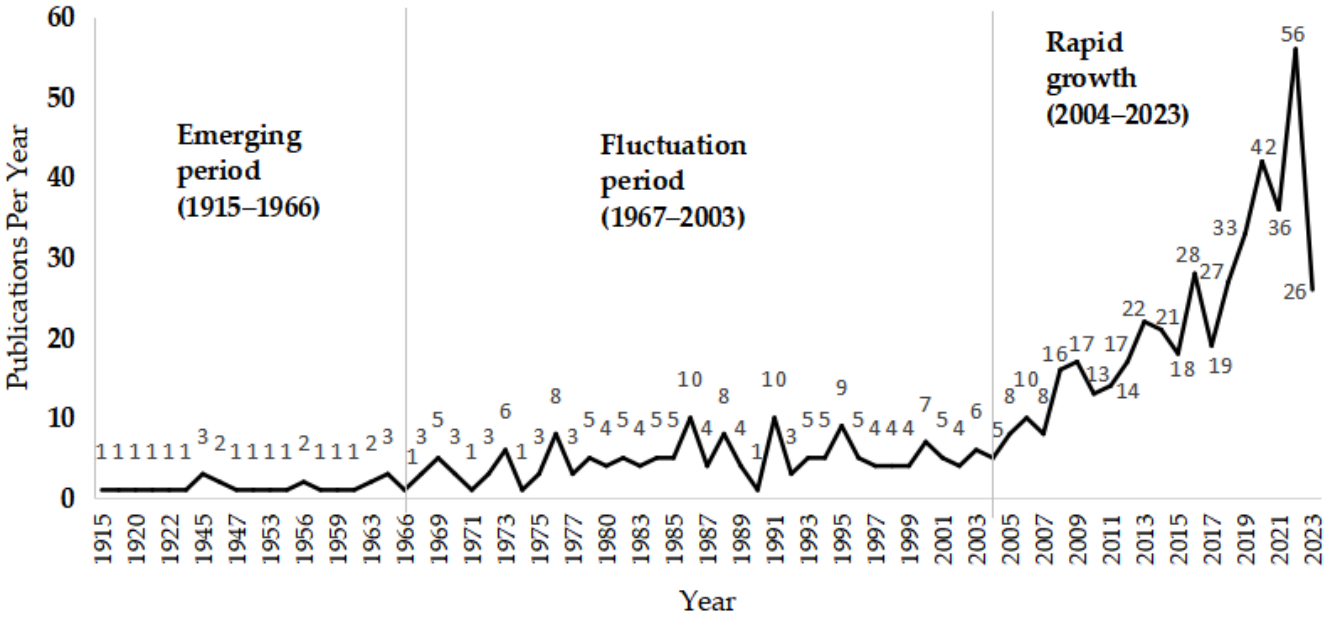


Figure 2. The trend of research on diseases and pests of rubber tree as revealed by annual publication. Three research periods can be identified based on their increasing trends: an emerging period from 1915 to 1966, a fluctuating period from 1967 to 2004, and a rapid growth period from 2005 to 2023. Data for the first half year of 2023 are represented in the figure as well.

The papers retrieved during data collection were published in 222 different journals. The most important sources on the topic of infectious diseases and pests of rubber tree were identified by applying Bradford's law [19]. These core journals were detected in zone 1 (Figure 3). This core, which implies the highly networked nature of this scientific field, consisted of fourteen journals: Plant Disease, Phytopathology, the European Journal of Plant Pathology, the Journal of Rubber Research, Plant Pathology, the European Journal of Forest Pathology, Physiological and Molecular Plant Pathology, Frontiers in Microbiology,

Pesquisa Agropecuaria Brasileira, Transactions of the British Mycological Society, Annals of Applied Biology, Forest Pathology, Mycopathologia, and Neotropical Entomology (Figure 3). Despite Plant Disease having the highest publication number, the European Journal of Forest Pathology, European Journal of Plant Pathology, Physiological and Molecular Plant Pathology, Mycopathologia and Phytopathology had higher local impacts.

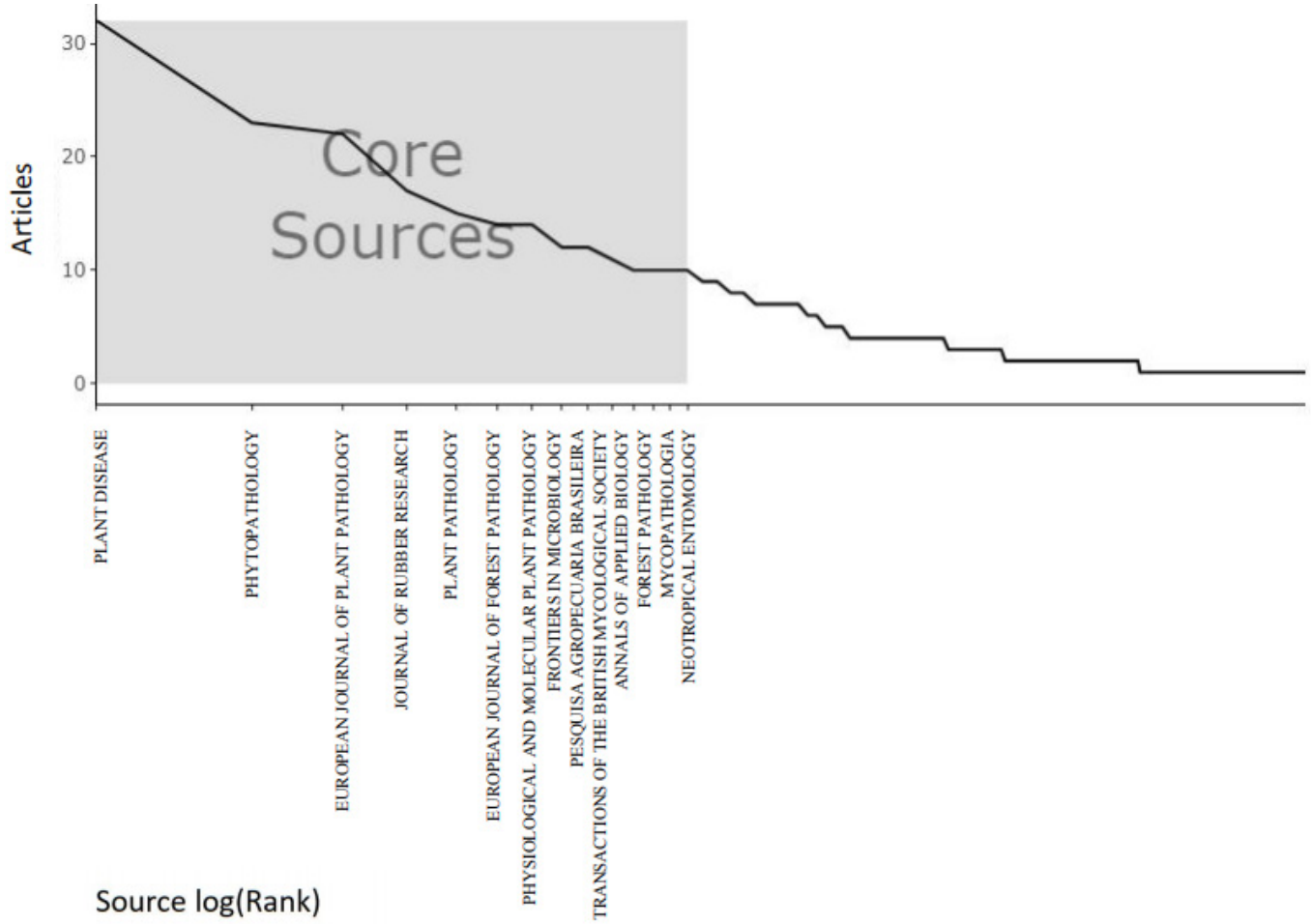


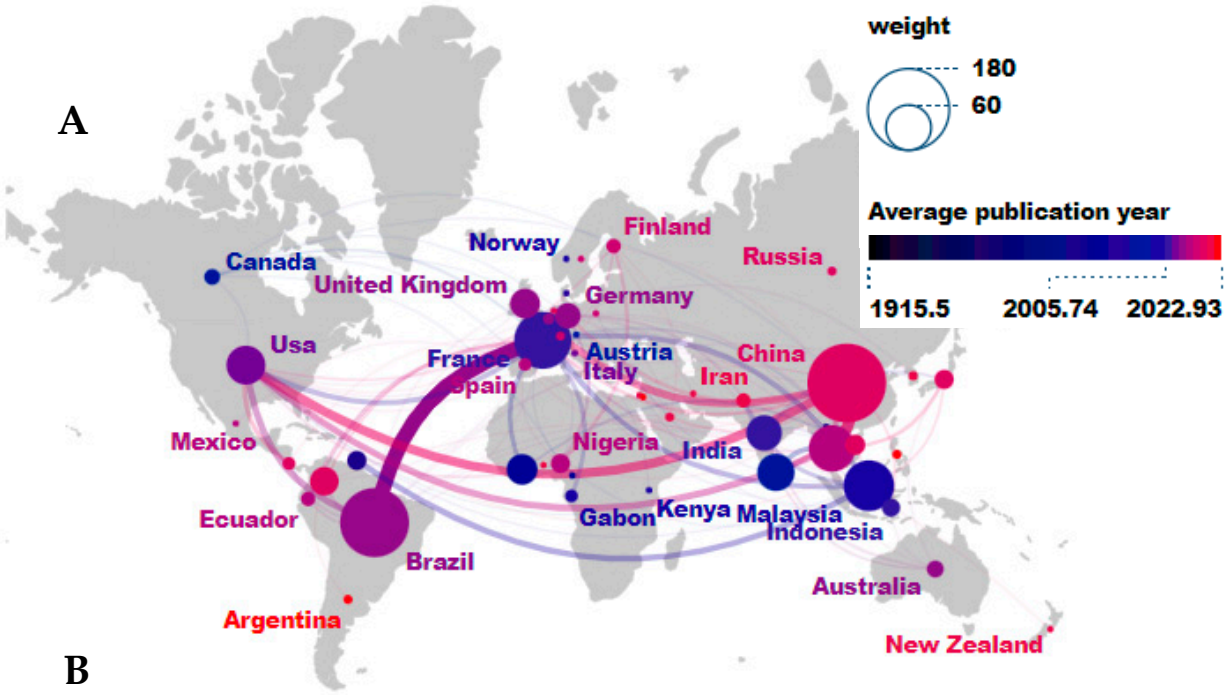
Figure 3. Most productive journals on diseases and pests of rubber tree in the literature. The shaded area includes the core sources in Zone 1 that were identified by applying Bradford's law.

3.2. Cooperation Analysis

Figure 4A shows the results after using VOSviewer and Scimago Graphica to analyze the distribution of publications by country/region [20,24,25]. The publications on diseases and pests of rubber tree are distributed among 51 countries/regions, as the map shows.

The country/region network visually shows the partnerships among countries/regions for publication. According to Figure 4A, cooperation is relatively close among the top five most productive countries of China, Brazil, France, Malaysia, and Thailand, as indicated by the thickness of the connecting line. Among the top five countries France seems to have the most co-authorship strength (total link strength = 65). France and Brazil have established the closest cooperation. China shows strong cooperation with Thailand, while the United States, and the United Kingdom. Malaysia cooperates with France, and the United States.

A



B

Position	Country/Region	Documents	Average publication year	Citations	Total link strength
1	China	152	2018	1109	56
2	Brazil	117	2011	1185	41
3	France	80	2006	1530	65
4	Malaysia	63	2006	484	30
5	Thailand	53	2014	589	39
6	USA	37	2009	742	46
7	Sri Lanka	34	1996	410	15
8	India	31	2006	221	13
9	Côte d'Ivoire	23	1988	310	6
10	United Kingdom	22	2011	257	23

Figure 4. Worldwide research on diseases and pests of rubber tree up to July 2023. (**A**): World map constructed using VOSviewer incorporated within Scimago Graphica; the picture depicts the contribution of each country based on publication count, average publication year, and total link strength. The bigger the circle of the country/region, the more documents were published by that country/region. The thickness of the connecting line represents Total Link Strength; a higher Total Link Strength indicates a higher co-authorship level. Countries with more co-authors are reflected by the Total Link Strength, indicating the total strength of the co-author links of a given country to other countries. (**B**): Publication information on the top ten countries was retrieved from VOSviewer software. Their position as ranked according to publication number is indicated as well.

The average publication year of documents from each county/region, indicated by the dark color from blue to red, spans from 1915 to 2022 (Figure 4A,B). Among the top five most productive countries, France and Malaysia seem to be the more active countries focusing on the research of the diseases and pests of rubber tree in the past, while in more recent years China and Thailand have become more active, with 2018 and 2014 as the average year of publication, respectively.

Among the top five most productive countries, China seems to have the shortest history of research on the diseases and pests of rubber tree (Figure 5). From the 1984 to 2008 the number of relevant studies started to accumulate slowly; however, after 2015 it increased sharply. Thailand, as a close partner of China, shows a similar trend, with the number of publications increasing after 2014. France and Brazil were more active in research on the diseases and pests of rubber tree in past years, from 1986 to 2007. Relevant research conducted by Malaysia has a long history, and the number of publications increased steadily until the year 1978; subsequently, the country has experienced a long period of steady publication until the year 2007, after which the rate of publication increases.

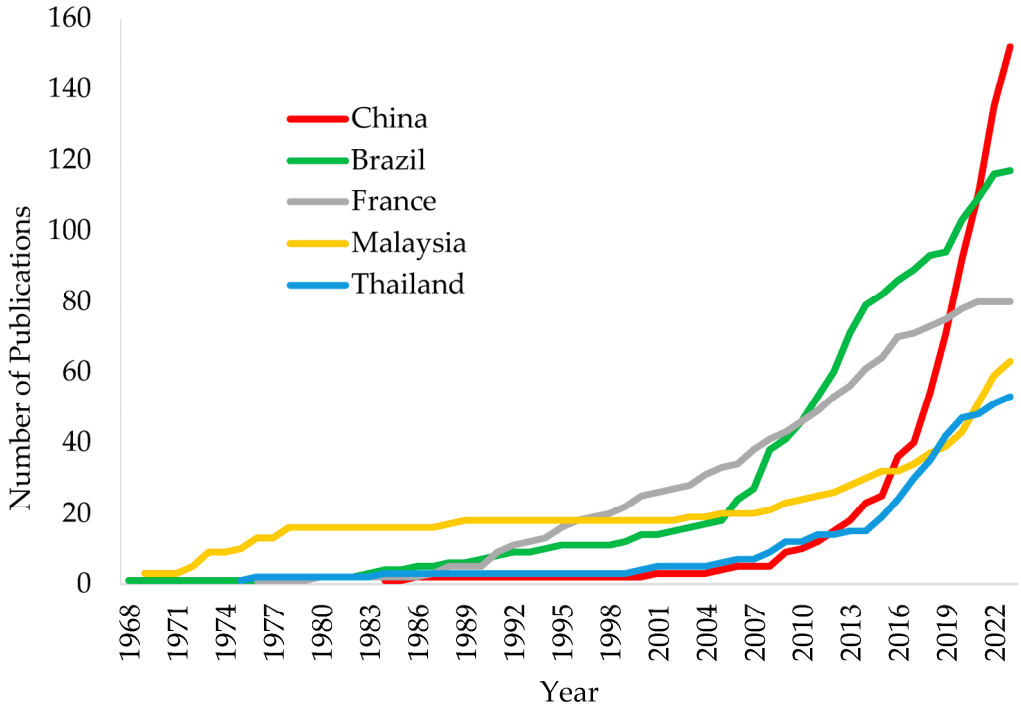


Figure 5. Number of documents by year in the research field on diseases and pests of rubber tree in the top five most productive countries.

According to Figure 6, the Chinese Academy of Tropical Agricultural Sciences and Hainan University in China have contributed the most to the research on diseases and pests of rubber tree. Rothamsted Research in England and Prince of Songkla University in Thailand were found to be their frequent partners (Figure 6). Among the top thirty most productive authors, researchers in China (Miao W, Liu W, Lin C, Huang G, Li X and He Q) form the largest cooperative group working on rubber diseases and pests (Figure 7). The Chinese Academy of Sciences is another important institute in China; they cooperate with Rubber Research Institute of Sri Lanka, Mae Fah Luang University, World Agrofor-estry Centre, Chiang Mai University, and University of Colombo (Figure 6). INRA and CIRAD, as the research organizations in France, have a long-term study of rubber diseases and pests; they mainly cooperate with important partners such as University of Montpellier, University Putra and the Rubber Research Institute of Malaysia. Among top thirty most productive authors, Nandris D (27 papers), Geiger JP (25 papers), Nicole M (25 papers), and Rio B (9 papers) mainly from France and Cote d'Ivoire have cooperated and published a high number of papers (Figures 6 and 7). Pujade-Renaud V, Garcia D and Le Guen V belong to another eminent research group in France (Figure 7). EMBRAPA, University Estadual Paulista and Plantacoes Michelin Bahia are the main research organizations in Brazil, and they have formed an important research group through cooperation to solve local issues regarding diseases and pests of rubber tree.

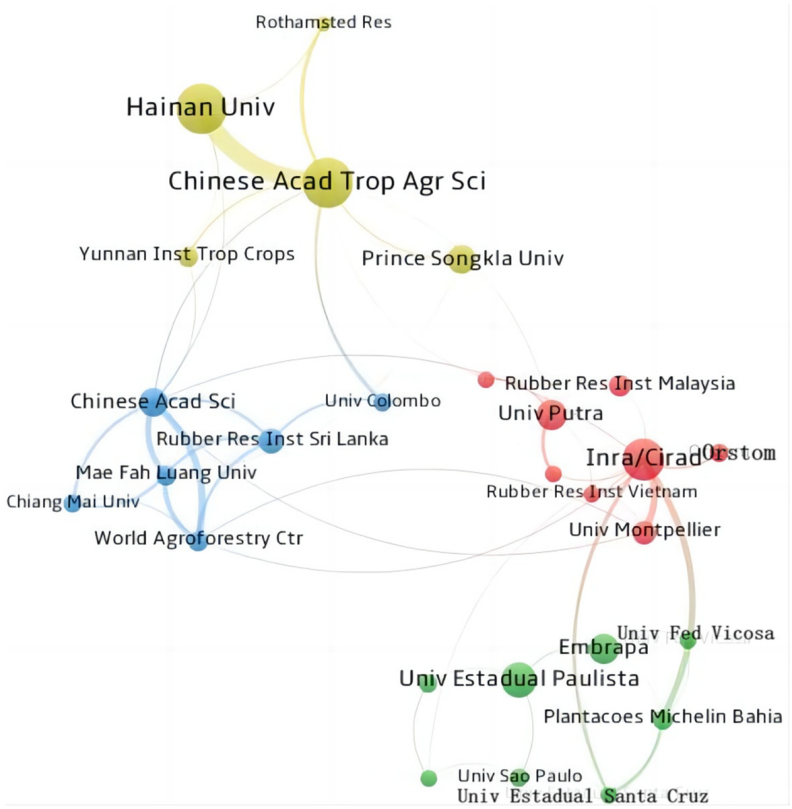


Figure 6. Collaboration network analysis of the top thirty most productive organizations in the domain of rubber diseases and pests. The size of the circle represents the number of documents resulting from cooperation, the thickness of the lines indicates the frequency of cooperation, and different colors present different clusters.

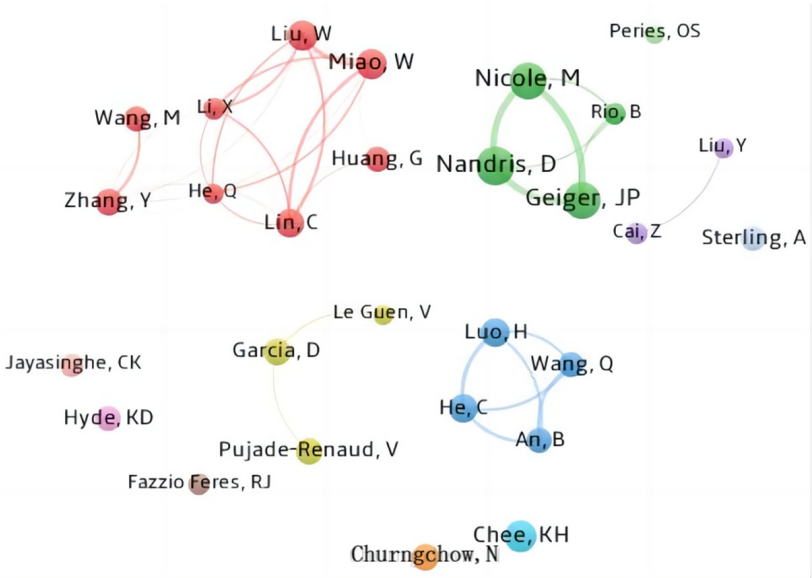


Figure 7. Collaboration network analysis of top thirty most productive authors in the domain of rubber diseases and pests. The size of the circles represents the number of documents resulting from author cooperation, the thickness of the lines indicates the frequency of cooperation, and the different colors represent different clusters.

3.3. Analysis of Themes and Trend Topics

Table 2 summarizes the main diseases and pests of rubber tree by calculating the number of relevant research publications. The top ten diseases and pests are listed in

Table 2: South American leaf blight (95 relevant documents with the average year of 2003) caused by *Pseudocercospora ulei*, root rot diseases (69 documents with the average year 2001) caused by *Rigidoporus microporus*, *Phellinus noxius*, *Armillaria* sp., etc., Colletotrichum leaf disease (72 documents with the average year 2011) caused by *Colletotrichum* species, emerging diseases that were first reported in different countries or regions (55 documents with the average year 2014), Phytophthora leaf fall caused by *Phytophthora* species (56 documents with the average year 1999), Corynespora leaf disease caused by *Corynespora cassiicola* (51 documents with the average year 2014), powdery mildew caused by *Oidium heveae*/*Erysiphe quercicola* (39 documents with the average year 2019), phytophagous mites (33 documents with the average year 2012) such as *Calacarus heveae*, *Tenuipalpus heveae*, *Eutetranychus banksi*, etc., subterranean termites (eight documents with the average year 2019) such as *Coptotermes curvignathus*, *Heterotermes tenuis*, *Coptotermes gestroi*, etc., and lace bug *Leptopharsa heveae* (seven documents with the average year 2009).

Table 2. Top ten diseases or pests of rubber tree based on relevant research publications. Each disease or pest is indicated with its main causal agents, document count, average publication year, and average citation count.

Position	Diseases or Pests	Causal Agents	Documents	Average Publication Year	Average Citation
1	South American leaf blight	*Pseudocercospora ulei*	95	2003	11.7
2	Colletotrichum leaf disease	*Colletotrichum acutatum*, *C. gloeosporioides*, *C. siamense*, etc.	72	2011	8.7
3	root rot diseases	*Rigidoporus microporus*, *Phellinus noxius*, *Armillaria* sp., etc.	69	2001	13.1
4	Phytophthora leaf fall	*Phytophthora meadii*, *P. palmivora*, *P. botryosa*, etc.	56	1999	10.8
5	emerging diseases	*Stemphylium lycopersici*, *Ceratocystis fimbriata*, *Euplatypus* sp., etc.	55	2014	5.7
6	Corynespora leaf disease	*Corynespora cassiicola*	51	2014	13.8
7	powdery mildew	*Oidium heveae*/*Erysiphe quercicola*, *E. necator*, etc.	39	2019	7.4
8	phytophagous mites	*Calacarus heveae*, *Tenuipalpus heveae*, *Eutetranychus banksi*, etc.	33	2012	9.5
9	subterranean termites	*Coptotermes curvignathus*, *Heterotermes tenuis*, *Coptotermes gestroi*, etc.	8	2019	1.5
10	lace bug	*Leptopharsa heveae*	7	2009	8.9

Three-field plots were used to visualize the connections between the selected fields (institutes, countries, and diseases/pests) and interpret their interrelationships using Sankey diagrams. Figure 8 represents flow diversion for the three fields of institutes (left), countries (middle), and diseases/pests (right) through different rectangles proportional to the relationship value. The three most productive institutes in China are: the Chinese Academy of Tropical Agricultural Sciences, Hainan University, and Chinese Academy of Sciences. The INRA and CIRAD organizations belong to France, while University Estadual Paulista and the EMBRAPA organization are from Brazil. University Putra and the Rubber Research Institute of Malaysia belong to Malaysia. Prince of Songkla University belongs to Thailand. The analysis indicates that, different countries have paid attention to different diseases/pests of rubber tree. The most important diseases or pests that attract the attention of China are Colletotrichum leaf disease (38 documents), powdery mildew (35 documents), Corynespora leaf disease (14 documents), and other emerging diseases (27 documents). France focuses attention on the diseases of South American leaf blight (27 documents), Corynespora leaf disease (13 documents), and root rot disease (13 documents). South American leaf blight (38 documents) and phytophagous mites (27 documents) are predominant in the research on diseases and pests of rubber tree in Brazil. Thailand has main interests in Phytophthora leaf fall (23 documents) and other emerging diseases (11 documents).



Figure 8. The relationships between institutes, countries, and diseases/pests. Three-field plots were used to visualize the connections for the three fields of institutes (**left**), countries (**middle**), and diseases/pests (**right**) through different rectangles proportional to the relationship value using the method of Sankey diagrams. The wider the connected lines are, the more documents are linked.

Table 3 displays the ten most cited documents in the research on diseases and pests of rubber tree. Vanparijs (1991), Pearce (1996), Shewry (1997), and Chye (1995) had the high citation numbers (258, 252, 147 and 59, respectively); they conducted research about plant defense and resistance. Galliano (1991) and Nandris (1987) studied the *Rigidoporus* species that causes root rot disease, with citation numbers of 132 and 58, respectively. Lieberei (2007) and Rocha (2011) conducted research about South American leaf blight of rubber tree, with citation numbers of 110 and 69, respectively. Rocha (2011) and Chaverri (2011) published research about biological control with high citations. Chen published "Plant disease recognition model based on improved YOLOv5" in 2022; in only one year this paper has gained a citation number of 55, which indicates that studying disease recognition using artificial intelligence has received great interest.

Table 3. Top ten most cited articles about rubber diseases and pests ordered by total citation number (TC).

#	Author(s)	Title	Journal	Publisher	Year	TC	TC/Y
1	Vanparijs et al.	Hevein-an antifungal protein from rubber-tree (*Hevea-brasiliensis*) latex	Planta	Springer Verlag	1991	258	8.1
2	Pearce et al.	Antimicrobial defences in the wood of living trees	New Phytologist	Wiley	1996	252	9.3
3	Shewry et al.	Plant proteins that confer resistance to pests and pathogens	Advances In Botanical Research	Elsevier Academic Press Inc.	1997	147	5.7
4	Galliano et al.	Lignin degradation by *Rigidoporus-lignosus* involves synergistic action of 2 oxidizing enzymes-mn peroxidase and laccase	Enzyme and Microbial Technology	Butterworth-Heinemann	1991	132	4.1
5	Lieberei, R	South American leaf blight of the rubber tree (*Hevea* spp.): New steps in plant domestication using physiological features and molecular markers	Annals of Botany	Oxford Univ Press	2007	110	6.9
6	Rocha et al.	Foliar endophytic fungi from *Hevea brasiliensis* and their antagonism on *Microcyclus ulei*	Fungal Diversity	Springer Verlag	2011	69	5.8
7	Chaverri et al.	*Trichoderma amazonicum*, a new endophytic species on *Hevea brasiliensis* and *H. guianensis* from the Amazon basin	Mycologia	Allen Press Inc.	2011	66	5.5
8	Chye et al.	β-1,3-glucanase is highly-expressed in laticifers of *Hevea-brasiliensis*	Plant Molecular Biology	Kluwer Academic Publ	1995	59	2.1
9	Nandris et al.	Root-rot diseases of rubber trees	Plant Disease	Amer Phytopathological Soc	1987	58	1.6
10	Chen et al.	Plant disease recognition model based on improved YOLOv5	Agronomy-Basel	MDPI	2022	55	55

The analysis of term co-occurrence performed in VOSviewer reflects the research trends and topics. The network visualization of the co-occurrences of all terms extracted from titles and abstracts is presented in Figure 9 [21]. The minimum number of a term was set at 5; of the 12,583 selected terms shown by the VOSviewer result, 1010 terms meet the threshold. These 1010 terms were then organized into five clusters represented by different colors with a total link strength of 497,010. Figure 9A comprises terms associated with 'pathogenicity/defense mechanisms' (red) in Cluster 1, 'disease control' (green) in Cluster 2, 'insect pests' (blue) in Cluster 3, 'disease identification' (yellow) in Cluster 4, and 'defense/resistance and clonal selection' (purple) in Cluster 5. The red color of Cluster 1 'pathogenecity/defense mechanisms' comprises associated terms such as *Colletotrichum gloeosporioide*, pathogenicity, infection, interaction, response, powdery mildew, disease resistance, *Phytophthora palmivora*, defense, salicylic acid, *Erysiphe quercicola*, molecular mechanism, defense mechanism, etc. The green color of Cluster 2 'disease control' is a cluster of associated terms such as control, fungicide, *Rigidoporus microporus*, biocontrol agent, antagonist, bioformulation, Arbuscular Mycorrhizal Fungi/AMF, etc. The blue color of Cluster 3 'insect pests' is associated with terms such as mite, Brazil, subterranean termite, *Calacarus heveae*, phytophagous mite, *Tenuipalpidae*, etc. The yellow color of Cluster 4 'disease identification' comprises associated terms such as identification, Corynespora leaf fall/CLD, cassiicolin gene, internal transcribed spacer, phylogenetic analysis. Finally, the purple color of Cluster 5 'defense/resistance and clonal selection' is a cluster of associated terms such as clone, resistance, South American leaf blight/SALB, *Microcyclus ulei/Pseudocercospora ulei*, genotype, susceptibility, resistant clone, durable resistance, etc.

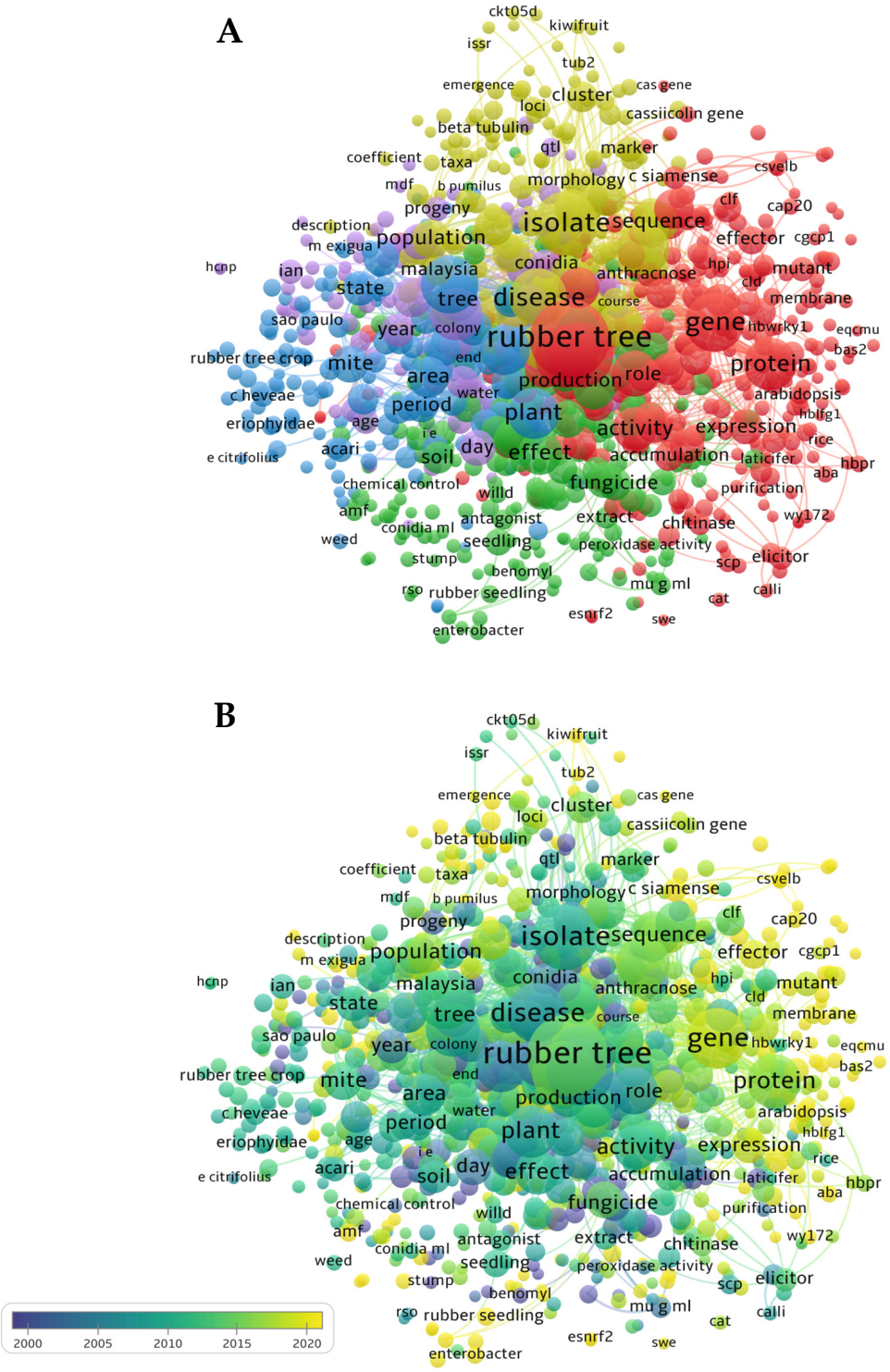


Figure 9. Co-occurrence network through the assessment of terms relating to diseases and pests of rubber tree. The term data retrieved from the title and abstracts of the publications were processed by the clustering algorithm of VOSviewer_1.6.18. Node size indicates occurrence of a term. (**A**): Network visualization of all terms, with five as the minimum number of occurrences of a term (weight: occurrences). Nodes with the same color in 9A represent a research cluster. (**B**): Overlap visualization of all terms, with five as the minimum number of occurrences of a term (weights: occurrences, score: average publications per year). Different colors indicate the evolution of terms over time according to the average publication year. Terms associated with recent research topics are colored yellow.

Based on the clustering-related articles and their occurrence terms (Figure 9A), several research conclusions can be drawn: studies on pathogenic mechanisms were primarily performed on the pathogens of *Colletotrichum* species, while the defense responses and resistant reactions were mainly investigated on the pathogen *O. heveae*/*E. quercicola* and *Phytophthora* species. Genetic diversity and pathogenic variety were frequently related with *C. cassiicola*; microbial biocontrol strategies were mainly investigated to combat root disease caused by *R. microporus*; resistant clone selection and breeding were usually related to the severity of South American leaf blight caused by *P. ulei*; and the application of predators as biocontrol agents for management of insects and other pests has been investigated (Figure 9B).

The evolution of topics over time is shown by the overlay visualization in Figure 9B. Terms colored purple represent earlier research topics, while yellow terms represent research topics in more recent years. According to Figure 9B, the recent hot topics indicated by the yellow-colored terms seem related to pathogenicity and defense mechanisms. These terms are: cell wall integrity, nucleus, plant immunity, actin cytoskeleton, hyphal tip, necrosis inducing activity, Eqcsep04187, Cgfim1, plant chitinase, rubber tree powdery mildew, *Colletotrichum siamense*, *E quercicola*, Hblfg1, rubber tree anthracnose, Cgnpg1, Hbmlo12, Cgnpg1, Hbmlo12, *C. australisinense*, necrotrophic effector, reactive oxygen species, CAS, ABA, Hbbik1, *Arabidopsis thaliana*, vegetative growth, etc.

3.4. Plant Defense and Resistance

To fight pathogenic fungi, the threatened rubber tree should be well-armed. The arsenal of molecular defense mechanisms deployed by rubber trees against diverse plant fungal pathogens comprises several main categories (Figure 10). Seven categories are summarized: genes associated with resistant proteins, transcription factors, protein kinases, proteases, protease inhibitor, glycosyl hydrolases, antioxidant enzymes, and other enzymes.

Genes associated with resistant proteins were mainly identified in rubber tree infected by *O. heveae*/*E. quercicola* and *C. gloeosporioides*: NPR1 (nonexpressor of PR1), PAD4 (phytoalexin deficient 4), EDS1 (enhanced disease susceptibility 1 protein), HbSGT1b (powdery mildew resistance-related gene), RPW8 (Resistance to powdery mildew 8), PMR4 (powdery mildew Resistant 4), HSP90 (heat shock protein), HbLFG1 (lifeguard protein), HbMLOs (mildew resistance locus O 1 to 34). Defense-related protein kinases were reported to be induced by *O. heveae*/*E. quercicola*, *C. gloeosporioides* and *R. microporus*: HbCDPK5 (calcium-dependent protein kinase), HbBIK1 (receptor-like kinase), STK (serine/threonine kinase), MAPK (mitogen-activated protein kinases). Three proteases were induced by *P. palmivora*/*Phytophthora* sp.: HbSPA (subtilisin-like serine protease), HbSPB (subtilisin-like serine protease) and HbSPC (subtilisin-like serine protease). Two protease inhibitors that were induced by *P. palmivora*/*Phytophthora* sp. were reported: HbCPI (cysteine protease inhibitor) and HbASI (alpha-amylase/subtilisin inhibitor). Antioxidant enzymes are associated with defense mechanisms: HbAPX (ascorbate peroxidase), HbCAT1 (catalase), HbCAT2 (catalase), HbCAT3 (catalase), PR9 (pathogenesis-related proteins/class IV peroxidase), POD (peroxidase), and SOD (superoxide dismutase). Glycosyl hydrolases were identified as: beta-glu (β-1,3-glucanase), GH18 (chitinase), HbCLP1 (GH19 chitinase-like protein), HbCLP2 (GH19 chitinase-like protein), HbGLU (β-glucosidase), HbPR1 (pathogenesis-related proteins/class I chitinase), PR3 (pathogenesis-related proteins/class I chitinase), PR5 (pathogenesis-related proteins/endochitinase), PR8 (pathogenesis-related proteins/endochitinase). Other defense related enzymes were identified: EXP (expansin), HbPAL (phenylalanine ammonia lyase), ACC oxidase (1-aminocyclopropane-1-carboxylate oxidase), AOC gene (Allene oxide cyclase), HNL (hydroxynitrile lyase), LIN (linamarase), HbNCED5 (9-cis-epoxycarotenoid dioxygenase 5), CAD (cinnamyl alcohol dehydrogenase), HbPPO (polyphenol oxidase). Transcription factors-HbEREBP1 (AP2/ERF transcription factors), HbEREBP2 (AP2/ERF transcription factors), HbMYB8-like (MYB transcription factor), HbWRKY1 (WRKY transcription factor), HbWRKY2 (WRKY transcription factor), HbWRKY40 (WRKY transcription factor) were all reported responsible for disease resis-

tance. Among these, WRKY transcription factor and MYB transcription factor were induced by fungal pathogens.

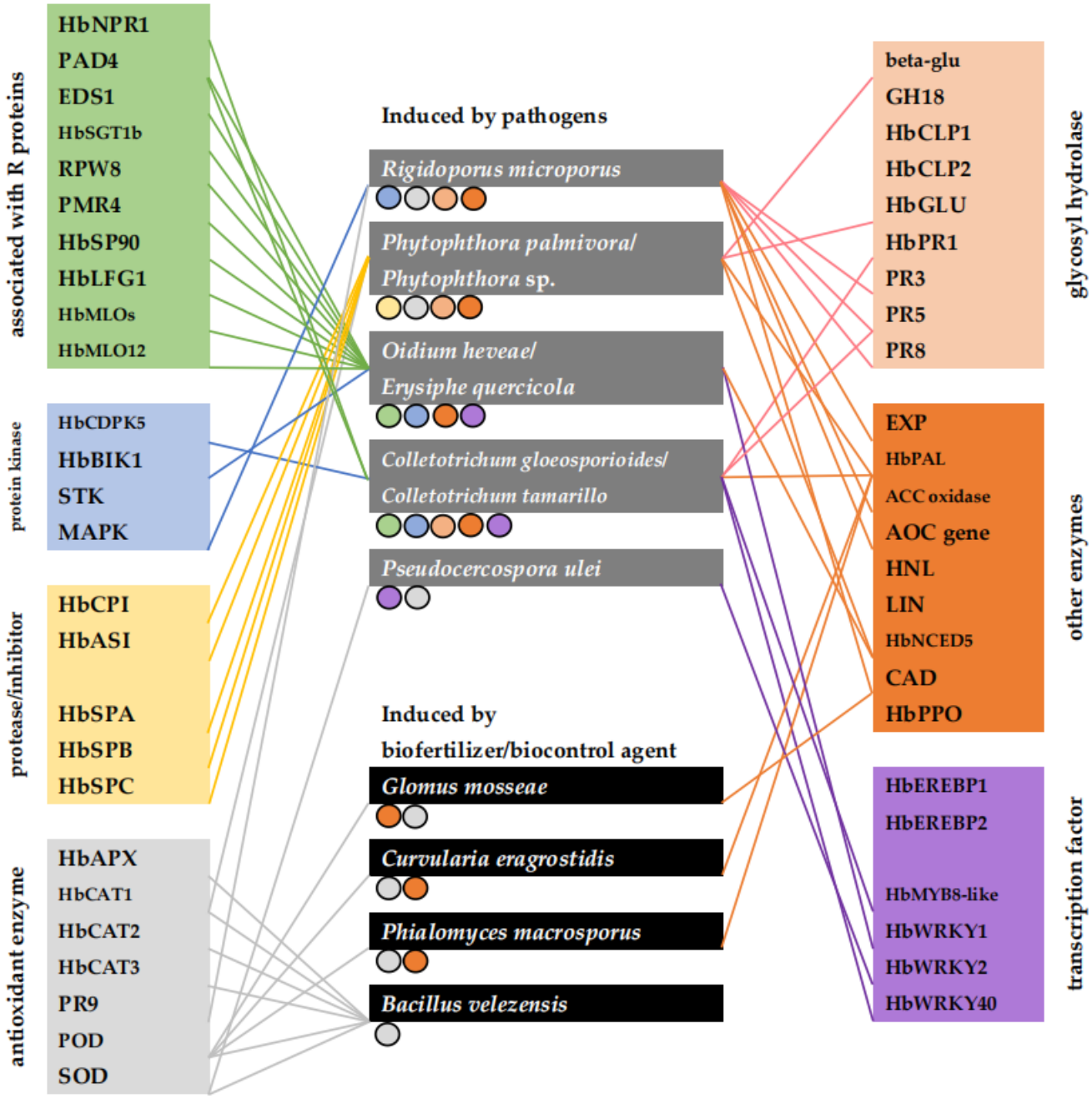


Figure 10. Genes studied for their involvement in defense mechanism of rubber tree induced by pathogens and biocontrol agents/biofertilizers [26–57]. Genes employed by rubber trees for disease resistance against pathogens or induced by selected biofertilizers/biocontrol agents were classified into seven groups exhibited by different colors. The genes and their targeted induced pathogens or biocontrol agents/biofertilizers are linked by lines with the same colors that were previously used to distinguish groups. In the middle column, the pathogens and biofertilizers/biocontrol agents are indicated by gray and black color, respectively.

The fungal pathogens that were frequently studied for their roles in the induction of rubber tree defense are listed in Figure 10, including *R. microporus*, *P. palmivora*/*Phytophthora* sp., *O. heveae*/*E. quercicola*, and *C. gloeosporioides*. Four categories of defense-related genes were expressed differentially in rubber tree when infected by *R. microporus*: protein kinases, gly-

cosyl hydrolases, antioxidant enzymes, and other enzymes. *P. palmivora*/*Phytophthora* sp. induced the differential expression of genes encoding glycosyl hydrolases, protease, protease inhibitors, and some enzymes. *O. heveae*/*E. quercicola* can particularly induce differential expression of an array of genes associated with the R proteins (PAD4, EDS1, HbSGT1b, RPW8, PMR4, HSP90, HbLFG1, HbMLOs). Defense responses of rubber tree were also explored when infected with *P. ulei*, though only a few defense related genes were identified. Because biocontrol agents and biofertilizers can promote plant immunity by inducing differential expression of defense related genes in rubber tree, the biocontrol agents of *Bacillus velezensis*, *Phialomyces macrosporus* and *Curvularia eragrostidis* and the biofertilizer *Glomus mosseae* have been studied for their potential roles in defense induction of rubber tree (Figure 10).

3.5. Biocontrol Candidates

Currently, there are at least 24 fungal species and seven bacteria showing mycoparasitic or antifungal activity on rubber fungal leaf pathogens (Figure 11). The fungal leaf pathogens that have been confronted by potential antagonists include *C. gloeosporioides*, *C. tamarilloi*, *C. cassiicola*, *E. quercicola*, *P. palmivora*, *P. meadii* and *P. ulei*. Antagonistic bacteria and fungi that were screened as biocontrol candidates for *C. gloeosporioides* include *Streptomyces deccanensis*, *S. badius*, *Bacillus subtilis*, *Aspergillus* sp., *Cophinforma atrovirens*, *Gliocladium* sp., *Gonatorrhodiella* sp., *Polyporales* sp., *Syncephalastrum* sp., *Trichocladium* sp., *Trichophyton* sp., *Clonostachys* sp., *Trichoderma* sp., *Curvularia eragrostidis*, and *Phialomyces macrosporus*. Fungal antagonists of *Dicyma pulvinata*, *Gibberella* sp., *Glomerella cingulate*, *Microsphaeropsis* sp., *Myrothecium* sp., *Pestalotiopsis* sp., and *Phomopsis* sp. were reported as biocontrol candidates for *P. ulei*. To combat *C. cassiicola*, *A. terreus*, *Diaporthe* sp., and *T. koningiopsis* have been screened. For controlling *E. quercicola*, *Ampelomyces* sp. and *B. velezensis* were reported. *Alcaligenes* sp., *B. amyloliquefaciens*, and *Pseudomonas aeruginosa* have been screened as biocontrol candidates for *P. meadii*.

Interestingly, researchers have tried to extract and characterize antifungal compounds from the plant extracts, and have found that such natural products can potentially be used as alternatives to harmful chemical fungicides (Figure 11). Plant extracts of *Ageratum conyzoides*, *Ocimum basilicum*, *Allium sativum*, *Jatropha curcas*, and *Vernonia amygdalina* have shown antifungal activity against *C. gloeosporioides*. Plant extracts of *Centrosema pubescene*, *Emilia coccinea*, and *Solanum torvum* inhibited fungal growth of *C. cassiicola*. Plant extracts of *Sargassum polycystum* were efficiently inhibited the growth of *P. palmivora* (Figure 11).

Currently, there are at least eleven known Ascomycota fungal species, four Basidiomycota fungal species, and nine bacteria that have shown mycoparasitic or antifungal activity against the rubber root pathogen *R. microporus* (Table 4). Among Ascomycota fungal species, the genera of *Trichoderma*/*Hypocrea* are frequently reported. Basidiomycota fungal species with biocontrol potential for *R. microporus* include *Lentinus squarrosulus*, *Cerrena meyenii*, *Gloeophyllum striatum*, and *Lentinus squarrosulus*. Among these bacteria antagonists, the genus of *Streptomyces* and *Pseudomonas* as well as species of *Bacillus subtilis*, *Burkholdcria* sp., and *Enterobacter* sp. have been screened as biocontrol candidates.

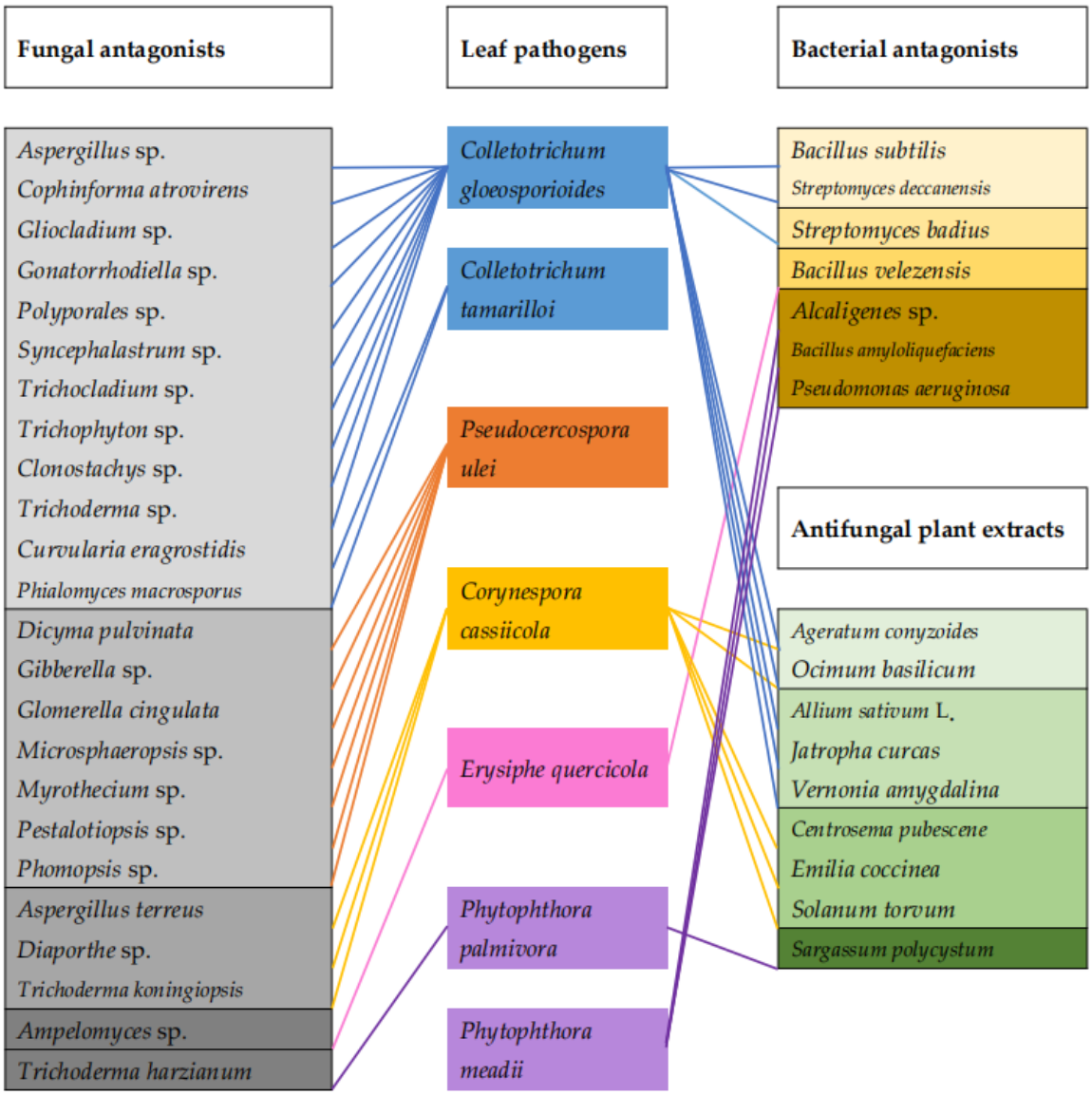


Figure 11. Biocontrol candidates considered in research on leaf pathogens of rubber tree [58–76]. The parasitism relationships of plant pathogens and their antagonists that have been studied are shown by lines with the same color of plant pathogens. The two side columns represent antagonists, while the middle column represents leaf fungal pathogens of rubber tree.

Table 4. Biocontrol agents/biofertilizers screened for root pathogen *R. microporus* of rubber tree.

Biocontrol Agents/Biofertilizers of *R. microporus*			
Bacteria	**Ascomycota**	**Basidiomycota**	**Glomeromycota**
Bacillus subtilis [77]	*Aspergillus terreus* [72]	*Lentinus squarrosulus* [77,78]	*Glomus mosseae* [79]
Burkholdcria sp. [12]	*Cladobotryum semicirculare* [80]	*Cerrena meyenii* [78]	*Glomus clarum* [81]
Enterobacter sp. [79]	*Chaetomiurn cupreum* [82]	*Gloeophyllum striatum* [78]	
Pseudomonas aeruginosa [83]	*Muscodor heveae* [84]	*Lentinus squarrosulus* [77,78]	
Pseudomonas sp. [12,85]	*Trichoderma* spp. [86]		
Streptomyces malaysiensis [87]	*T. asperellum* [88]		
Streptomyces ahygroscopicus [87]	*Trichoderma spirale* [88]		
Streptomyces sp. [89]	*Trichoderma harzianum* [90]		
Streptomyces sioyaensis [89]	*Hypocrea lixii* [91]		
	Hypocrea virens [91]		
	Hypocrea jecorina [91]		

4. Discussion

4.1. Disease Focuses of the Top Five Most Productive Countries

The research distribution among countries/regions and their cooperation network reveals research perspectives, changes, and progress worldwide. Analyzing the main diseases and pests of rubber tree in countries and the research trend through the years can help to better understand the research hot spots, core issues, and topics of interest during a specific period.

4.1.1. China

In China, the most dangerous diseases of rubber tree are anthracnose and powdery mildew caused by *Colletotrichum* complex species and *E. quercicola*, respectively; *E. quercicola*, an obligate biotrophic pathogen, is supposed to be the only causal agent of powdery mildew of rubber tree in Hainan, Yunnan, and Guangdong province of China [5,92]. Hence, rubber leaf diseases of anthracnose and powdery mildew have become two main research directions for Chinese researchers (Figure 8). China has published 152 documents during recent investigations, with 2018 as the average publication year (Figure 4B). The Chinese Academy of Tropical Agricultural Sciences, Hainan University, and Chinese Academy of Sciences are the dominant institutes in publishing researches on the diseases and pests of rubber tree in China (Figures 6 and 8). Several productive Chinese researchers from these institutes have formed a number of large research groups focusing on Colletotrichum anthracnose and powdery mildew in rubber tree (Figures 6 and 7) [93,94]. The analysis of term co-occurrence performed on VOSviewer reflects the research trends and topics [21]. The 'pathogenicity/defense mechanisms' cluster comprises associated terms such as *Colletotrichum gloeosporioide*, pathogenicity, infection, interaction, response, powdery mildew, disease resistance, defense, salicylic acid, *Erysiphe quercicola*, *Phytophthora palmivora*, and defense mechanism (Figure 9A). Based on the clustering of related articles and their occurrence terms, this cluster primarily illustrates studies on pathogenic mechanisms performed on the pathogens of *Colletotrichum* species, while defense responses and resistant reactions were mainly investigated on the pathogen *O. heveae*/*E. quercicola* and *Phytophthora* species.

The pathogenicity mechanisms of *Colletotrichum* species in rubber tree have been identified and subjected to molecular analysis by Chinese researchers. Genes responsible for vegetative growth, conidiation, germination, appressorium formation, plant penetration, disease infection, and spreading inside plant cells have been characterized and investigated. Among the genes responsible for lifecycle completion and pathogenicity of *Colletotrichum* species, transcription factors CgAzf1 and CgHSF1 play a role through activation of melanin biosynthesis and regulation of gene expression [95,96]. NOXs and CgFim1 have functions in pathogenicity through remodeling and organization of the actin cytoskeleton [97,98]. Genes including CgPBS(2), CsAtf1, Pbs2, and Hog1 are responsible for fungal sensitivity to osmotic stress and fungal resistance to fungicides [99,100]. Genes such as CgRGS1, CgMFS1, and CgOPT2 are essential for oxidative stress during the infection process [101–103]. Effectors such as BAS2, CgCP1, and CgNLP1 secreted from *Colletotrichum* species enable the pathogen to successfully penetrate into the host tissue of the rubber tree [54,104,105].

The causal agent of powdery mildew was first considered to be *O. heveae*, though later research suggested *E. quercicoia* based on morpho-molecular studies [106]. The response of rubber tree to infection by *E. quercicola* has been explored by Chinese researchers; for example, differential expression of defense-related genes induced by *E. quercicoia* in rubber tree has been identified and characterized (Figure 6).

4.1.2. Brazil

Brazil has published the second largest amount of research regarding diseases and pests of rubber tree (117 documents), with 2011 the average publication year (Figure 4). Important phytophagous mites (especially *Tenuipalpus heveae* and *Calacarus heveae*) inhabiting rubber plantations and the fungal pathogen *P. ulei* that causes South American leaf

blight have aroused great attention on the part of Brazilian researchers (Figure 8). EMBRPA and University Estadual Paulista cooperate in trying to find solutions to solve local issues involving diseases and pests of rubber tree (Figure 6) [56,107].

In Brazil, South American leaf blight is a significant limiting factor in natural rubber production [3]. One solution relies on genetic manipulation to grow rubber varieties with resistance or tolerance to the disease [108]. Another solution is developing new rubber plantations in areas where climatic conditions are adverse to growth of the fungus *P. ulei*, where the disease spreads less easily [109]. In the cluster 'defense/resistance and clonal selection', terms including South American leaf blight/SALB and *Microcyclus ulei*/*Pseudocercospora ulei* were clustered with terms such as resistance, genotype, susceptibility, resistant clone, and durable resistance. This indicates that resistant clone selection and breeding are usually conducted for control of long-term South American leaf blight disease (Figure 9A).

The phytophagous mites *T. heveae* and *C. heveae* are important rubber tree pests in Brazil [110]. The Cluster 'insect pests' shows high frequency of terms such as mite, Brazil, subterranean termite, *Calacarus heveae*, phytophagous mite, Tenuipalpidae, predatory mite, temperature, rubber tree clone, distribution, and climatic factor. This cluster indicates that control of rubber pests has been widely investigated. To strengthen epidemic information tracking and monitoring, the population dynamics of mites on different rubber tree clones have been frequently investigated in Brazil [111]. To control the pests in an environmentally friendly way, the employment of predatory mites can be developed as a good biocontrol strategy [112]. The use of resistant genetic material in rubber clones has been explored as well, with resistant progenies with tolerance to or no preference for at least one mite species selected for pest control [113–115].

4.1.3. France, Malaysia and Thailand

Root rot diseases result in major economic loss to rubber production in all rubber tree growing countries [100]. Root rot diseases were the focus of attention of French researchers besides South American leaf blight and Corynespora leaf fall disease (Figure 8). One eminent group formed by Nandris D, Geiger JP, Nicole M, and Rio B have focused their attention on white root disease in rubber tree; one of the relevant papers by Nandris et al., is among the top ten most cited articles in the field of rubber diseases (Figure 7 and Table 3) [116]. In close cooperation with French organizations, Malaysia has published at least seventeen documents in research on root rot diseases caused by *Rigidoporus* sp. (Figures 6 and 8). Biocontrol agents for white root rot disease have been extensively selected from among bacteria and fungi (Table 4). Current disease control measures rely on chemical fungicides, which cause environmental pollution; thus, the utilization of biocontrol microorganisms seems to be an attractive alternative in controlling root rot diseases of rubber tree [12].

The necrotrophic fungus *C. cassiicola* causes disease across a wide range of hosts [117]. Corynespora leaf disease of rubber tree causes severe yield loss of rubber in African and Asian plantations; however, its influence is negligible in South America [74,118]. The studies on Corynespora leaf fall disease of rubber tree were almost evenly distributed among China, France, and Malaysia (Figure 8). The severity of leaf infection caused by different *C. cassiicola* isolates varies significantly [119,120]. Most research about *C. cassiicola* was closely related to keywords such as "morphology", "phylogenetic analysis", and "pathogenicity test" (Figure 9A); indicating that investigation of Corynespora leaf disease of rubber tree usually starts with understanding the genetic variety of *C. cassiicola*. Cassiicolins, as small secreted glycoproteins, are important phytotoxic effectors in *C. cassiicola*; the diversity of the cassiicolin-encoding genes seems to be responsible for pathogen virulence during interaction with rubber hosts [121].

Certain aggressive oomycete *Phytophthora* sp. induce black stripe and abnormal leaf fall in rubber trees [122]. Thailand researchers have published the highest number of papers on investigation of *Phytophthora* sp. (Figure 8). The study of defense mechanisms induced by *Phytophthora* fungi is one of these research hot spots (Figure 10). To gain insights into the

interaction mechanism between *Phytophthora* pathogens and *H. brasiliensis*, defense-related genes such as subtilisin-like serine proteases have been identified [48].

4.2. Induced Expression of Defense as a Hallmark of Resistance to Rubber Plant Fungal Pathogens

Screening and characterization of resistant rubber clones for disease management is environmentally friendly, though it is always time consuming [123]. However, most clones of *H. brasiliensis* that produce high yield of latex are more disease-susceptible. In order to obtain resistant rubber clones, resistant evaluations and pathogenicity tests have been conducted for a wide range of rubber pathogens on various *H. brasiliensis* clones. For the tested clones, defense related genes have been searched and identified. We have summarized defense-related genes that were differentially expressed induced by fungal pathogens or biocontrol agents in Figure 10. Structural polymorphism and differential expression in disease resistance genes reveal great potential for plant genetic management and engineering [124]. Further in-depth studies into these defense-related genes would help to better understand the resistant properties and their regulation mechanisms; most importantly, they would provide a promising gene pool for further diversification of the rubber genetic resource. The achieved knowledge would be beneficial to engineer resistance and then further devise natural control strategies for protection of Hevea seedlings from pests and pathogens.

How rubber trees establish successful defense boundaries against the causal agent of powdery mildew remains largely unknown in spite of many investigations being conducted. PAD4 (Phytoalexin Deficient 4) and EDS1 (Enhanced Disease Susceptibility 1) have been identified as effector-triggered immunity proteins. The callose synthase PMR4 (Powdery Mildew Resistant 4) helps to prevent *O. heveae* infection [27]. The RPW8s (Resistance to Powdery Mildew 8), as proven broad-spectrum resistance genes, can be induced to combat the powdery mildew fungi; among them, the HbRPW8-protein, one of the four HbRPW8s (a, b, c, and d) in rubber tree, increases its expression significantly during powdery mildew infection. It has been localized to the plasma membrane, where it induces plant immune responses [31]. The Lifeguard protein HbLFG1, as a negative regulator, facilitates the infection of *E. quercicola* by suppressing plant immunity [33]. HbMlo12 might confer susceptibility to powdery mildew [36]. HbBIK1 encodes a receptor-like cytoplasmic kinase functions in plant immune response to *O. heveae* infection [38]. HbNCED5 encodes 9-cis-epoxycarotenoid dioxygenase, and is involved in biosynthesis of abscisic acid (ABA) in plant cell chloroplasts; its localization can be disturbed by the effector protein EqCSEP01276 secreted from powdery mildew fungus [45]. *O. heveae*'s attack strongly enhances the transcriptional accumulation of the WRKY transcriptional factor HbWRKY1 in latex [55]. The roles played by defense-related genes can enlighten researchers in developing new rubber tree varieties with powdery mildew resistance through reverse genetics.

In order to find molecular solutions to increase the resistance and tolerance of rubber tree to abnormal leaf fall disease and white root rot disease, defense-related genes responding to the infection of *Phytophthora* species or *R. microporus* have been identified and characterized (Figure 10). The cysteine proteinase inhibitor HbCPI might be involved in defense mechanism, as its accumulation in leaf, latex, or young seed can be induced by *P. palmivora* infection [43]. HbASI, an alpha-amylase/subtilisin inhibitor, was induced in the rubber leaves by *P. palmivora* inoculation; HbASI could inhibit *P. palmivora*'s mycelium growth [44]. Increases of enzyme activities of phenylalanine ammonia lyase (PAL), polyphenol oxidase, peroxidase and catalase have been found to be essential means of reducing disease severity in rubber trees infected by *P. palmivora* [32]. The β-1,3-glucanase (beta-glu) enzyme was proven to be pathogenesis-related in *P. meadii*-tolerant clones of rubber tree [28]. Genes involved in sensing and signal transduction (ACC oxidase, AOC, and MAPK) and cell wall modification (PAL and expansin) as well as pathogenesis-related proteins (PR1, PR3, PR5, PR8, and PR9) play a critical role in successfully combatting white rot disease caused by *R. microporus* [34]. Defense-related genes provide rational gene candidates for genetic transformation and pro-defense genetic breeding.

4.3. Biocontrol Candidates: Mechanisms and Applications

Currently, screening of biocontrol agents has been widely conducted among rhizospheric soilborne fungi, endophytes, phylloplane microflora, and even extracts from specific plants. The disease control capacity of the selected biocontrol agents have revealed great eco-friendly natural control potential. Biocontrol mechanisms for rubber tree place emphasis on four aspects: antibiotic ability, pathogen cell wall degradation, plant resistance induction, and plant growth promotion.

4.3.1. Antibiotic Ability

Antifungal metabolites such as lipopeptides of strain *B. velezensis* HN-2, rhamnolipid from *P. aeruginosa* USM-AR2, and dinactin produced by *S. badius* strain gz-8- inhibit pathogen growth [52,62,63,83]. Lipopeptides induced morphological changes in *E. quercicola* spores [52]. A complex lipopeptides mixture secreted by local *B. subtilis* Czk1 contained members of bacillomycin, iturin, surfactin, and fengycin; these ingredients showed high antifungal activities against a broad range of fungal pathogens such as *C. gloeosporioides*, *Ganoderma pseudoferreum*, *R. lignosus*, *Phellinus noxius*, *Sphaerostilbe repens*, and *Helicobasidium compactum* [59]. Dinactin of *S. badius* gz-8 caused impairments in the conidiogenesis and conidia germination of *C. gloeosporioides* [62]. Metabolites of salicylic acid, siderophore, and HCN produced by endophytes *Pseudomonas* sp. and *Bacillus* sp. showed antifungal activity against *C. cassiicola* and *P. meadii* [85]. Phenazine-1-Carboxylic Acid (PCA) enabled the endophyte *Alcaligenes* sp. EIL-2 to successfully inhibit *P. meadii* [63].

4.3.2. Cell Wall Degradation

Because cell wall-degrading enzymes promote mycoparasitic behavior of mycelia when killing fungal pathogens, the candidates of biocontrol agents selected for disease control of rubber tree have mainly been characterized for their ability to produce these enzymes. *S. ahygroscopicus* LBR 14, *S. malaysiensis* Lac 17, and *S. seoulensis* Lac 19 were selected for white root disease control based on their ability to produce cellulase, chitinase, and catalase [87]. *T. atroviride* P1 produced a new chitinase (Ech30); the architecture of its active site resembles that of endochitinases from *H. brasiliensis*, such as hevamine [125]. Chitinolytic microorganisms from the genus *Streptomyces*, *Burkholdcria*, and *Pseudomonas* degraded mycelium of *R. microporus* by hydrolyzing the cell wall using chitinase [12]. β-1,3-glucanase secreted by endophyte *B. amyloliquefaciens* showed great catalytic potential and strong antagonism toward *P. meadii* [64]. *D. pulvinata* produced hydrolytic enzymes to hydrolyze *P. ulei* [126].

4.3.3. Induced Plant Resistance

B. velezensis secreted HN-2 elicitors into rubber leaves; these elicitors enhance plant immunity by suppressing HbLFG1 and elevating the activity of antioxidant enzymes such as ascorbate peroxidase, peroxidase, catalase, and superoxide dismutase [52]. When treated with the saprobic fungal antagonists *Curvularia eragrostidis* and *Phialomyces macrosporus*, phenylalanine ammonia lyase activity was higher in rubber tree, which enhanced plant resistance against the anthracnose pathogen *C. tamarilloi* [39]. Class III peroxidases are associated with defense responses to *P. meadii*; their activity in rubber leaves can be induced by *Alcaligenes* sp. [127]. After treatment with extract of the brown seaweed *Sargassum polycystum*, peroxidase, catalase, beta-1,3-glucanase, salicylic acid, and scopoletin were increased in rubber tree; as a result, its application was able to enhance plant resistance and reduce leaf fall disease caused by *P. palmivora* [73].

4.3.4. Plant Growth Promotion

The bacterial endophytes *Pseudomonas* sp. and *Bacillus* sp. improved the photosynthetic efficiency of rubber seedlings and starch granule accumulation in root tissues [85]. The local isolate *S. sioyaensis* has been applied for plant growth promotion and sustainable

control of root invasion in rubber tree, based partially on its ability to produce indole-3-acetic acid [89]. Endophytic isolates of the genera *Bacillus* and *Enterobacter* from healthy *H. brasiliensis* produced indole-3-acetic acid and ACC deaminase, indicating that these isolates have great potential for plant growth promotion [128]. *Enterobacter* sp. and the arbuscular mycorrhizal fungus *G. mosseae* conferred growth promotion on rubber seedlings when applied for control of white root rot disease [79].

5. Conclusions

This bibliometric analysis has aimed to provide an overview of the diseases and pests of rubber tree across countries worldwide. The top five countries and their representative of diseases and pests have been the focus of this study. Molecular mechanisms employed by pathogens were primarily investigated for *Colletotrichum* species, particularly in the recent years by Chinese research teams. Resistant clone selection and breeding were usually used to deal with the severity of South American leaf blight caused by *P. ulei*, with most research being conducted by France and Brazil. Plant resistant properties and associated defense responses were recently examined in relation to *O. heveae*, *R. microporus*, and *P. palmivora*. Phytotoxicity of *C. cassiicola* and its genetic diversity has been the focus of great attention on the part of China, France, and Malaysia. Molecular mechanisms used by antagonistic biocontrol agents have been explored by targeting them to specific rubber tree pathogens. Integrated control methods have a long investigation history in Brazil for management of both insects and other pests.

The severity of rubber tree disease brought about by fungal pathogens, especially in tropical regions, increases the urgency of finding efficient control solutions. In order to propose and develop natural control projects, a solid molecular-level knowledge of the resistance properties and defense mechanisms in different clones of the rubber tree and the pathogenicity mechanisms of their virulent representatives is needed. One of natural control strategy relies on the selection of biocontrol strains and the investigation of their antagonistic mechanisms. Another strategy for developing new rubber tree varieties involves the exploring of resistance related genes. Our literature investigation suggests that several antagonists have been selected as biocontrol agents and a number of defense related genes in rubber tree have been characterized in recent years. The limited research based on resistant genes for genetic breeding and antagonistic mechanisms for biocontrol candidate selection reveals great potential for developing future natural disease control strategies.

Author Contributions: Conceptualization, G.H., L.C. and L.X.; methodology, L.C. and L.X.; Data analysis, L.C., X.L., L.X., Y.W. and Y.F.; Investigation, L.C. and X.L.; writing—original draft preparation, L.C. and L.X.; writing—review and editing, G.H., L.C. and Y.F.; supervision, G.H. and L.C.; project administration, G.H. and L.C.; funding acquisition, G.H. and L.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Central Public-interest Scientific Institution Basal Research Fund, grant number "1630072021010-JY".

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

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